

# CCNA Lab Study Notes

Static Routing & Centralized DHCP across a 5-router hub-and-spoke topology (Suren Network lab, captured 9 Sep 2026)

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This document walks through every screenshot from the lab in order, explaining what each command output shows and how it fits into the overall design. The goal is to read it as a self-contained study guide: topology first, then how traffic is routed between sites, then how each site receives its IP addresses automatically via DHCP, and finally how the client PCs confirm that the design actually works.

## What this lab demonstrates

- A **hub-and-spoke static routing** design: R2 is the hub; R1, R3, R4, and R5 are spokes, each attached to one LAN (Sales, HR, MGMT, Production).
- A **centralized DHCP server** running on R3, which hands out addresses for all four LANs even though most of those LANs sit behind other routers.
- Verification technique: reading `show ip interface brief` and `show ip route` on every router, then confirming the result on the Windows clients with `ipconfig`.

# 1. Network Topology

Five routers form a star around **R2**. Each spoke router connects one LAN of PCs to the core. The labelled subnets on each link are the point-to-point router-to-router networks; the purple boxes are the LANs where end devices (Win-1 to Win-4) live.

FIGURE 1 | Overall topology (Physical view, Packet Tracer / GNS3)

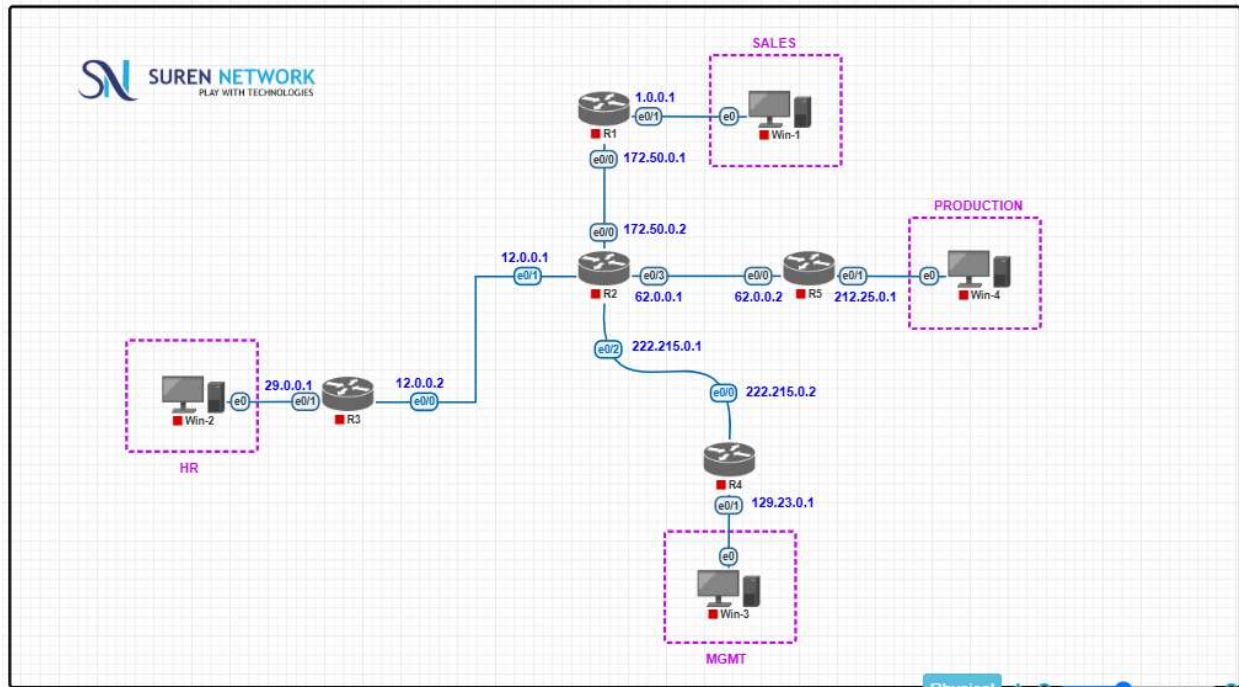


Figure 1 — R2 is the hub. R1 (Sales), R3 (HR), R4 (MGMT), and R5 (Production) are the spokes, each with one attached LAN.

## 1.1 Addressing plan

Reading the IP labels in Figure 1 gives the full addressing table for the lab:

| Link / LAN     | Router | Interface | IP address         | Remote end            |
|----------------|--------|-----------|--------------------|-----------------------|
| R1 ↔ R2        | R1     | e0/0      | 172.50.0.1         | R2 e0/0 = 172.50.0.2  |
| R1 ↔ Sales LAN | R1     | e0/1      | 1.0.0.1 (gateway)  | Win-1                 |
| R2 ↔ R1        | R2     | e0/0      | 172.50.0.2         | R1                    |
| R2 ↔ R3        | R2     | e0/1      | 12.0.0.1           | R3 e0/0 = 12.0.0.2    |
| R2 ↔ R4        | R2     | e0/2      | 222.215.0.1        | R4 e0/0 = 222.215.0.2 |
| R2 ↔ R5        | R2     | e0/3      | 62.0.0.1           | R5 e0/0 = 62.0.0.2    |
| R3 ↔ R2        | R3     | e0/0      | 12.0.0.2           | R2                    |
| R3 ↔ HR LAN    | R3     | e0/1      | 29.0.0.1 (gateway) | Win-2                 |

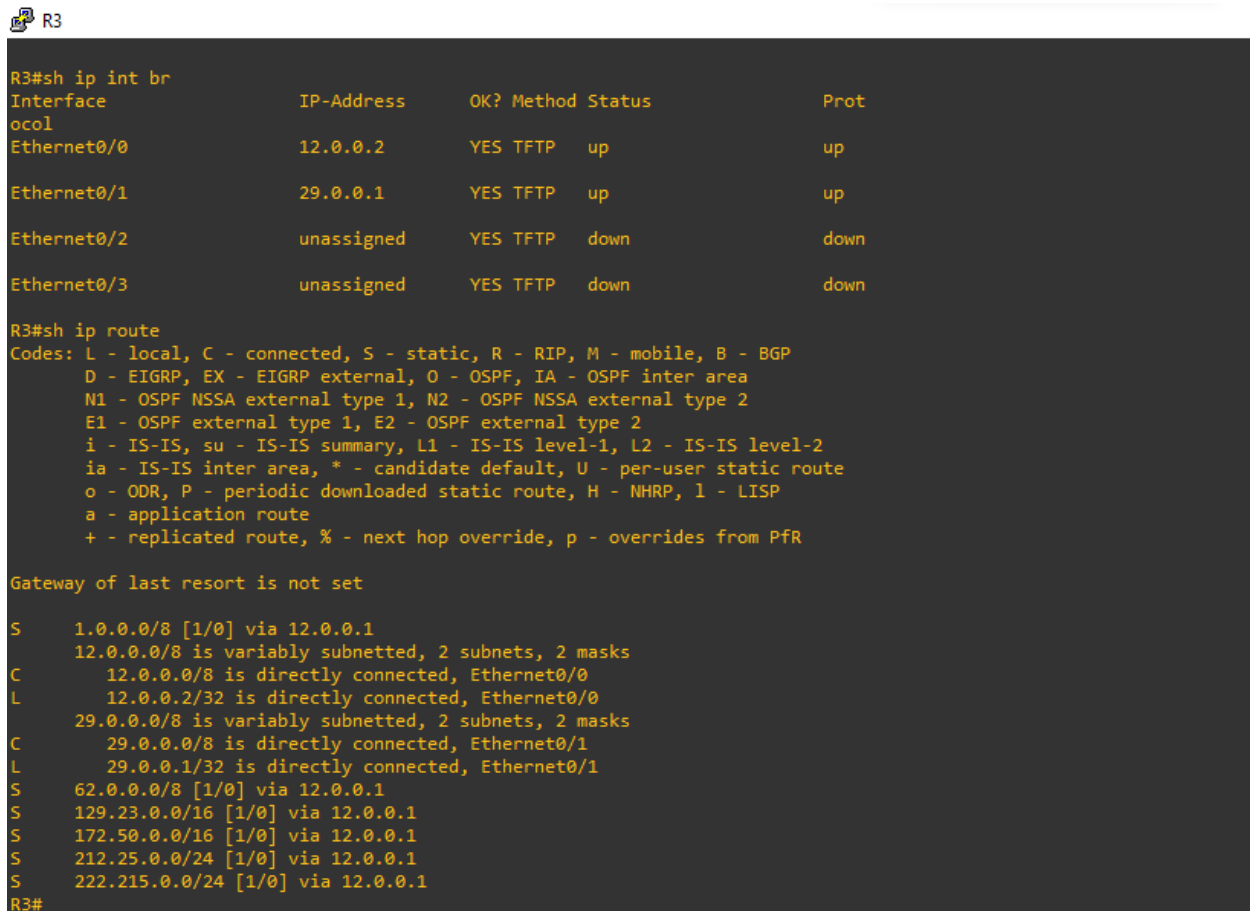
| Link / LAN          | Router | Interface | IP address           | Remote end |
|---------------------|--------|-----------|----------------------|------------|
| R4 ↔ R2             | R4     | e0/0      | 222.215.0.2          | R2         |
| R4 ↔ MGMT LAN       | R4     | e0/1      | 129.23.0.1 (gateway) | Win-3      |
| R5 ↔ R2             | R5     | e0/0      | 62.0.0.2             | R2         |
| R5 ↔ Production LAN | R5     | e0/1      | 212.25.0.1 (gateway) | Win-4      |

Notice every LAN uses a different major network (1.0.0.0, 29.0.0.0, 129.23.0.0, 212.25.0.0) and every inter-router link is its own subnet too. Nothing overlaps, which is exactly what makes static routing manageable here — with only five routers, a full mesh of static routes is still readable.

## 2. Static Routing: How Each Router Reaches Every Other LAN

None of these routers are running a dynamic routing protocol — every route below is a manually configured static route (code **S** in the routing table), and Gateway of last resort is not set on all five, confirming there's no default route either. The pattern is simple once you see it: **every spoke has a static route pointing at R2 for every network it doesn't own, and R2 has a static route pointing at each spoke for that spoke's LAN.**

FIGURE 2 | R3 — interfaces and routing table



```
R3
R3#sh ip int br
Interface IP-Address OK? Method Status Prot
ocol
Ethernet0/0 12.0.0.2 YES TFTP up up
Ethernet0/1 29.0.0.1 YES TFTP up up
Ethernet0/2 unassigned YES TFTP down down
Ethernet0/3 unassigned YES TFTP down down

R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S 1.0.0.0/8 [1/0] via 12.0.0.1
  12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 12.0.0.0/8 is directly connected, Ethernet0/0
L 12.0.0.2/32 is directly connected, Ethernet0/0
  29.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 29.0.0.0/8 is directly connected, Ethernet0/1
L 29.0.0.1/32 is directly connected, Ethernet0/1
S 62.0.0.0/8 [1/0] via 12.0.0.1
S 129.23.0.0/16 [1/0] via 12.0.0.1
S 172.50.0.0/16 [1/0] via 12.0.0.1
S 212.25.0.0/24 [1/0] via 12.0.0.1
S 222.215.0.0/24 [1/0] via 12.0.0.1
R3#
```

Figure 2 — R3's own subnets (12.0.0.0/8 and 29.0.0.0/8) are Connected (C); everything else (1.0.0.0/8, 62.0.0.0/8, 129.23.0.0/16, 172.50.0.0/16, 212.25.0.0/24, 222.215.0.0/24) is a static route via 12.0.0.1 — R3's only way out is through R2.

### 2.1 R2 — the hub

R2 has four active interfaces (e0/0–e0/3), one toward each spoke, so it sees four directly-connected networks. For the LANs that sit behind each spoke, R2 needs one static route per remote LAN, pointing at that spoke's WAN address:

FIGURE 3 | R2 — interfaces and routing table

```
R2#
R2#sh ip int br
Interface                IP-Address      OK? Method Status      Prot
-----
Ethernet0/0              172.50.0.2      YES TFTP   up          up
Ethernet0/1              12.0.0.1        YES TFTP   up          up
Ethernet0/2              222.215.0.1     YES TFTP   up          up
Ethernet0/3              62.0.0.1        YES TFTP   up          up

R2#
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    1.0.0.0/8 [1/0] via 172.50.0.1
    12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
      C    12.0.0.0/8 is directly connected, Ethernet0/1
      L    12.0.0.1/32 is directly connected, Ethernet0/1
    S    29.0.0.0/8 [1/0] via 12.0.0.2
    62.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
      C    62.0.0.0/8 is directly connected, Ethernet0/3
      L    62.0.0.1/32 is directly connected, Ethernet0/3
    S    129.23.0.0/16 [1/0] via 222.215.0.2
    172.50.0.0/16 is variably subnetted, 2 subnets, 2 masks
      C    172.50.0.0/16 is directly connected, Ethernet0/0
      L    172.50.0.2/32 is directly connected, Ethernet0/0
    S    212.25.0.0/24 [1/0] via 62.0.0.2
    222.215.0.0/24 is variably subnetted, 2 subnets, 2 masks
      C    222.215.0.0/24 is directly connected, Ethernet0/2
      L    222.215.0.1/32 is directly connected, Ethernet0/2
R2#
```

Figure 3 — R2's four static routes: 1.0.0.0/8 via 172.50.0.1 (R1/Sales), 29.0.0.0/8 via 12.0.0.2 (R3/HR), 129.23.0.0/16 via 222.215.0.2 (R4/MGMT), 212.25.0.0/24 via 62.0.0.2 (R5/Production).

## 2.2 R1, R4, R5 — the other spokes

R1, R4, and R5 all follow the same pattern as R3: two connected networks (their WAN link to R2 and their own LAN), and a static route toward **every other** network in the lab, all with R2's interface as the next hop.

FIGURE 4 | R1 — interfaces and routing table

```
R1
R1(config)#exit
R1#
*Sep  9 17:36:13.328: %SYS-5-CONFIG_I: Configured from console by console
R1#
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       1.0.0.0/8 is directly connected, Ethernet0/1
L       1.0.0.1/32 is directly connected, Ethernet0/1
S       12.0.0.0/8 [1/0] via 172.50.0.2
S       29.0.0.0/8 [1/0] via 172.50.0.2
S       62.0.0.0/8 [1/0] via 172.50.0.2
S       129.23.0.0/16 [1/0] via 172.50.0.2
        172.50.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.50.0.0/16 is directly connected, Ethernet0/0
L       172.50.0.1/32 is directly connected, Ethernet0/0
S       212.25.0.0/24 [1/0] via 172.50.0.2
S       222.215.0.0/24 [1/0] via 172.50.0.2
R1#
R1#sh ip int br

```

| Interface   | IP-Address | OK? | Method | Status | Protocol |
|-------------|------------|-----|--------|--------|----------|
| Ethernet0/0 | 172.50.0.1 | YES | manual | up     | up       |
| Ethernet0/1 | 1.0.0.1    | YES | TFTP   | up     | up       |
| Ethernet0/2 | unassigned | YES | TFTP   | down   | down     |
| Ethernet0/3 | unassigned | YES | TFTP   | down   | down     |

```
R1#
```

Figure 4 — R1 connects Sales (1.0.0.0/8) and reaches all five remote networks via 172.50.0.2 (R2). Note interface e0/0 shows method 'manual' (statically configured), while e0/1 shows 'TFTP' (loaded from a saved startup-config).

FIGURE 5 | R4 — interfaces and routing table

 R4

```
R4(config-if)#exit
R4(config)#exit
R4#sh
*Sep  9 17:38:04.003: %SYS-5-CONFIG_I: Configured from console by console
R4#sh ip int br
Interface                IP-Address      OK? Method Status        Prot
-----
o0c1
Ethernet0/0              222.215.0.2     YES TFTP   up            up
Ethernet0/1              129.23.0.1      YES TFTP   up            up
Ethernet0/2              unassigned      YES TFTP   down          down
Ethernet0/3              unassigned      YES TFTP   down          down

R4#
R4#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    1.0.0.0/8 [1/0] via 222.215.0.1
S    12.0.0.0/8 [1/0] via 222.215.0.1
S    29.0.0.0/8 [1/0] via 222.215.0.1
S    62.0.0.0/8 [1/0] via 222.215.0.1
     129.23.0.0/16 is variably subnetted, 2 subnets, 2 masks
C     129.23.0.0/16 is directly connected, Ethernet0/1
L     129.23.0.1/32 is directly connected, Ethernet0/1
S    172.50.0.0/16 [1/0] via 222.215.0.1
S    212.25.0.0/24 [1/0] via 222.215.0.1
     222.215.0.0/24 is variably subnetted, 2 subnets, 2 masks
C     222.215.0.0/24 is directly connected, Ethernet0/0
L     222.215.0.2/32 is directly connected, Ethernet0/0
R4#
R4#
```

Figure 5 — R4 connects MGMT (129.23.0.0/16) and reaches everything else via 222.215.0.1 (R2).

FIGURE 6 | R5 — interfaces and routing table

```
R5
R5(config-if)#exit
R5(config)#exit
R5#
R5#
*Sep  9 17:38:53.768: %SYS-5-CONFIG_I: Configured from console by console
R5#sh ip int br

```

| Interface   | IP-Address | OK? | Method | Status | Protocol |
|-------------|------------|-----|--------|--------|----------|
| Ethernet0/0 | 62.0.0.2   | YES | TFTP   | up     | up       |
| Ethernet0/1 | 212.25.0.1 | YES | TFTP   | up     | up       |
| Ethernet0/2 | unassigned | YES | TFTP   | down   | down     |
| Ethernet0/3 | unassigned | YES | TFTP   | down   | down     |

```
R5#
R5#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    1.0.0.0/8 [1/0] via 62.0.0.1
S    12.0.0.0/8 [1/0] via 62.0.0.1
S    29.0.0.0/8 [1/0] via 62.0.0.1
      62.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      62.0.0.0/8 is directly connected, Ethernet0/0
L      62.0.0.2/32 is directly connected, Ethernet0/0
S    129.23.0.0/16 [1/0] via 62.0.0.1
S    172.50.0.0/16 [1/0] via 62.0.0.1
      212.25.0.0/24 is variably subnetted, 2 subnets, 2 masks
C      212.25.0.0/24 is directly connected, Ethernet0/1
L      212.25.0.1/32 is directly connected, Ethernet0/1
S    222.215.0.0/24 [1/0] via 62.0.0.1
R5#
```

Figure 6 — R5 connects Production (212.25.0.0/24) and reaches everything else via 62.0.0.1 (R2).

## 2.3 Why this matters for the exam

- A route with an **administrative distance shown as [1/0]** in brackets confirms it's a static route (AD 1) with a metric of 0 — you'll see [1/0] next to every S entry in these tables.
- Static routing here is **not** redundant: each spoke has exactly one path out (through R2), so if R2 goes down, every spoke loses connectivity to every other spoke and LAN.
- This is the classic trade-off tested on the exam: static routes are simple and predictable for small, stable topologies like this one, but don't scale or self-heal the way EIGRP/OSPF would.



### 3. Centralized DHCP Service on R3

Instead of configuring a DHCP pool on every router, this design puts **all four DHCP pools on R3** — including pools for LANs that are physically attached to R1, R4, and R5, not R3 itself. That only works because R1, R4, and R5 must be configured with `ip helper-address 12.0.0.2` (or R2's relevant address) on their LAN-facing interfaces, which converts each PC's DHCP broadcast into a unicast request R3 can answer across the routed network. This is the standard CCNA "DHCP relay" scenario.

**FIGURE 7 | R3 — DHCP pool summary (sh ip dhcp pool)**

```
R3#
R3#sh ip dhcp pool

Pool HR :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 16777214
  Leased addresses                : 1
  Pending event                  : none
  1 subnet is currently in the pool :
  Current index      IP address range      Leased addresses
  29.0.0.52          29.0.0.1 - 29.255.255.254  1

Pool MGMT :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 65534
  Leased addresses                : 1
  Pending event                  : none
  1 subnet is currently in the pool :
  Current index      IP address range      Leased addresses
  129.23.0.57        129.23.0.1 - 129.23.255.254  1

Pool Sales :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 16777214
  Leased addresses                : 1
  Pending event                  : none
  1 subnet is currently in the pool :
  Current index      IP address range      Leased addresses
  1.0.0.58           1.0.0.1 - 1.255.255.254  1

Pool Pro :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses                : 0
  Pending event                  : none
  1 subnet is currently in the pool :
  Current index      IP address range      Leased addresses
  212.25.0.1         212.25.0.1 - 212.25.0.254  0

R3#
R3#
R3#
R3#
```

Figure 7 — Four pools: HR (29.0.0.0/8), MGMT (129.23.0.0/16), Sales (1.0.0.0/8), and Pro (212.25.0.0/24). Each pool's size comes from its classful default mask — Cisco's `ip dhcp pool` command uses a classful mask unless you configure the network statement with an explicit /24, /16, etc.

#### 3.1 Reading the pool sizes

| Pool name | Network | Default mask used | Total addresses | Serves LAN |
|-----------|---------|-------------------|-----------------|------------|
| Sales     | 1.0.0.0 | /8 (Class A)      | 16,777,214      | R1 — Win-1 |

| Pool name | Network    | Default mask used | Total addresses | Serves LAN |
|-----------|------------|-------------------|-----------------|------------|
| HR        | 29.0.0.0   | /8 (Class A)      | 16,777,214      | R3 — Win-2 |
| Pro       | 212.25.0.0 | /24 (Class C)     | 254             | R5 — Win-4 |
| MGMT      | 129.23.0.0 | /16 (Class B)     | 65,534          | R4 — Win-3 |

The client subnet masks confirm this later in Figures 10–13: Win-1 and Win-2 both show 255.0.0.0, Win-3 shows 255.255.0.0, and Win-4 shows 255.255.255.0 — exactly matching the classful defaults above.

**FIGURE 8 | R3 — DHCP address bindings (sh ip dhcp binding)**

```

R3#
R3#
R3#
R3#sh ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/          Lease expiration    Type
                   Hardware address/
                   User name
1.0.0.57            0150.b7bb.06ac.00   Sep 10 2026 11:02 PM Automatic
29.0.0.51           0150.cd1b.06ad.00   Sep 10 2026 10:15 PM Automatic
129.23.0.56         0150.ce12.06ae.00   Sep 10 2026 10:56 PM Automatic
R3#

```

Figure 8 — At the moment this was captured, three leases were active: 1.0.0.57 (Sales), 29.0.0.51 (HR), and 129.23.0.56 (MGMT).

One detail worth noticing for troubleshooting practice: the Pool summary in Figure 7 shows the **Pro** pool with 0 leased addresses at capture time, and it doesn't appear in the binding table in Figure 8 either — yet the Win-4 client later shows a leased address of 212.25.0.71 (Figure 13). That's not a config error; it simply means the Production PC requested (or renewed) its lease after these two particular screenshots were taken. On a real exam task or in your own lab, always re-run `show ip dhcp binding` right before you check the client, rather than relying on an older capture.

## 4. Client-Side Verification (ipconfig)

The last step in any DHCP/routing lab is proving it end-to-end from the client. Each Windows PC below is a DHCP client, and each result should match the pool it's meant to draw from and the LAN gateway shown back in Figure 1.

FIGURE 9 | Win-1 (Sales LAN) — ipconfig

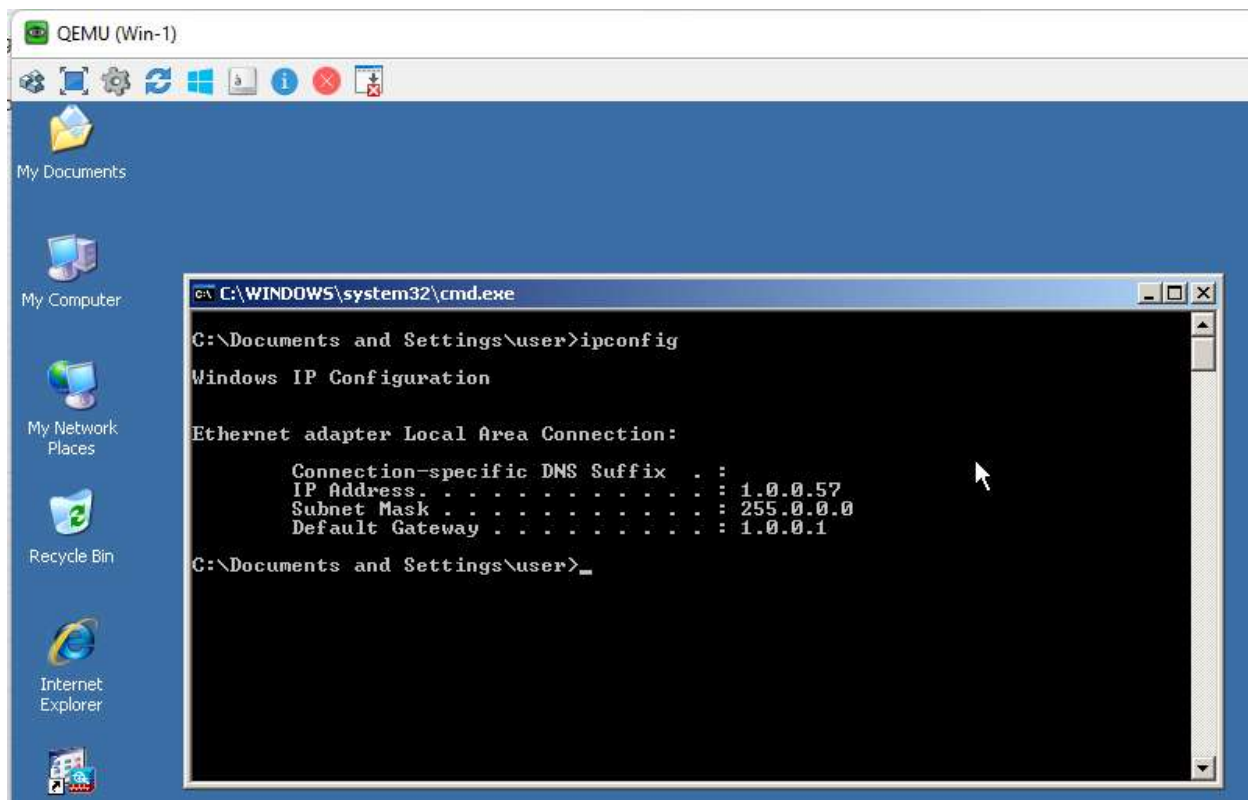


Figure 9 — 1.0.0.57 / 255.0.0.0, gateway 1.0.0.1. Matches the Sales pool and R1's e0/1 address.

FIGURE 10 | Win-2 (HR LAN) — ipconfig

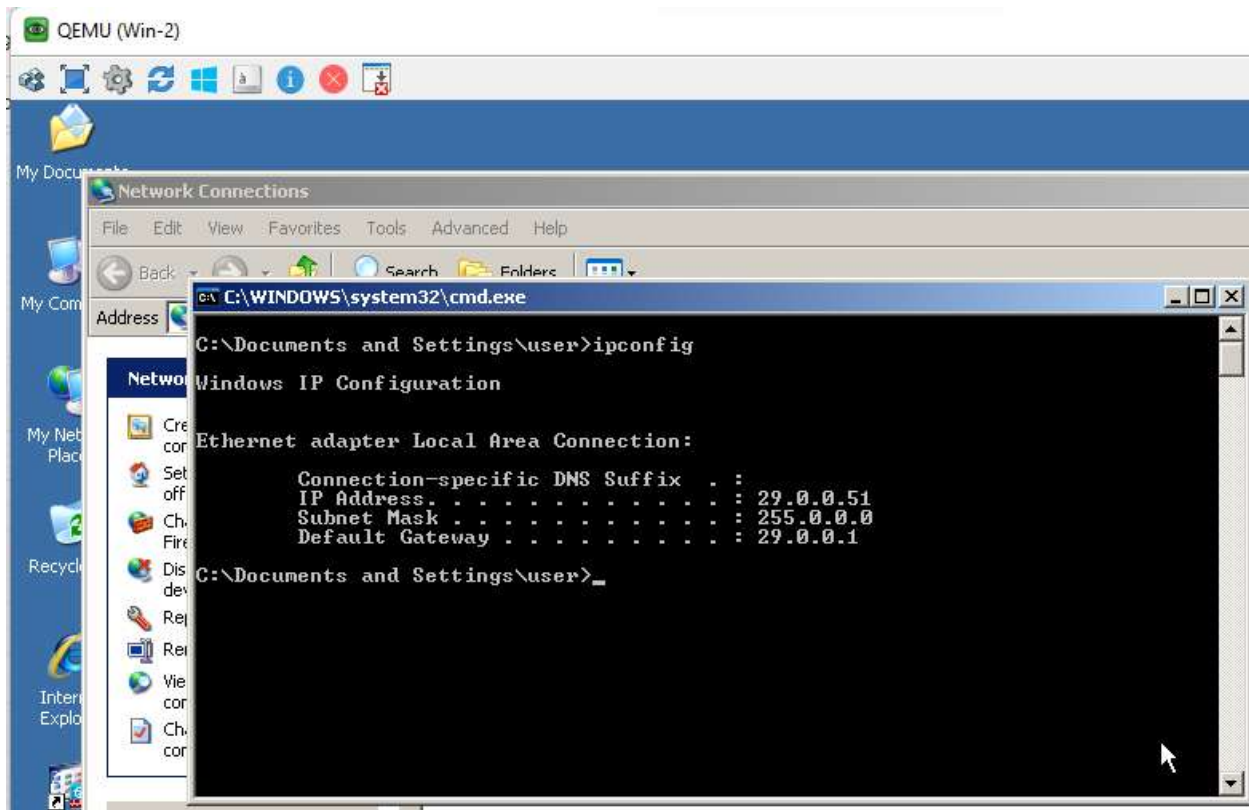


Figure 10 — 29.0.0.51 / 255.0.0.0, gateway 29.0.0.1. Matches the HR pool and R3's e0/1 address.

FIGURE 11 | Win-3 (MGMT LAN) — ipconfig

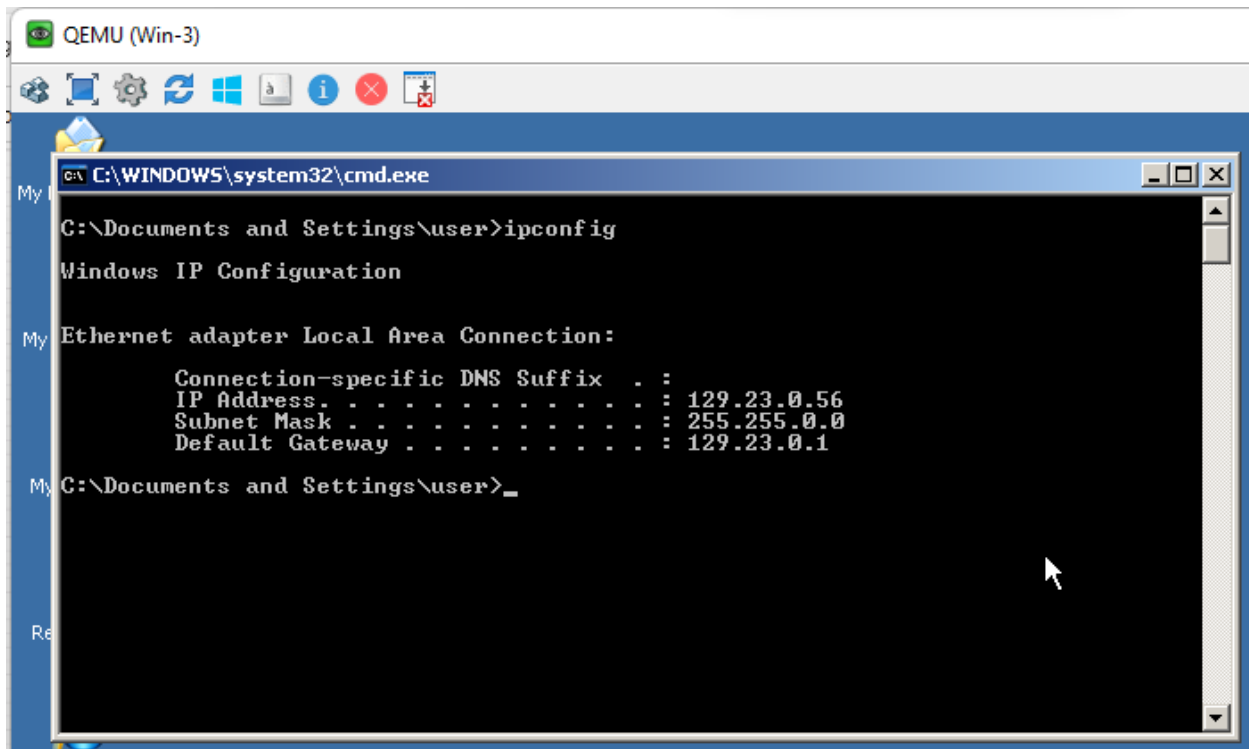


Figure 11 — 129.23.0.56 / 255.255.0.0, gateway 129.23.0.1. Matches the MGMT pool and R4's e0/1 address.

FIGURE 12 | Win-4 (Production LAN) — ipconfig

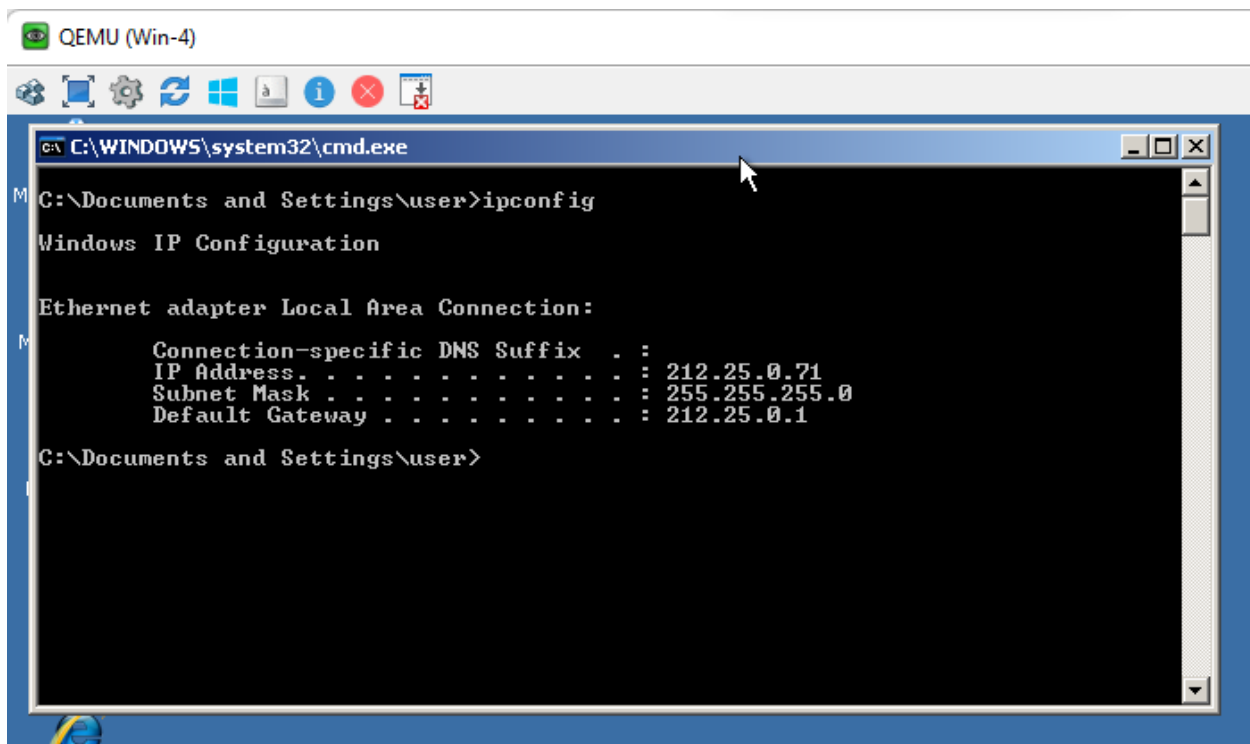


Figure 12 — 212.25.0.71 / 255.255.255.0, gateway 212.25.0.1. Matches the Pro pool and R5's e0/1 address — this is the lease that hadn't yet shown up in Figures 7–8.

Every client received an address from the correct classful pool and a gateway equal to its local router's LAN interface — confirming that both the static routing between routers and the DHCP relay from each LAN back to R3 are working correctly.

## 5. Key CCNA Takeaways

- **show ip interface brief** — fastest way to check which interfaces are up/up and their assigned IPs before troubleshooting anything else.
- **show ip route** — 'C' = connected, 'L' = local (the router's own interface address as a /32), 'S' = static. [AD/metric] in brackets tells you how the route was learned and how it will be preferred against other sources.
- **Hub-and-spoke static routing** scales by hand only for small topologies — every spoke needs one route per remote network, and the hub needs one route per spoke LAN.
- **ip dhcp pool** uses the classful default mask for the network unless you explicitly narrow it — always check 'Total addresses' in `show ip dhcp pool` to catch an oversized or undersized scope.
- **DHCP relay (ip helper-address)** is what lets a single centralized DHCP server (here, R3) serve LANs that live behind completely different routers.
- **show ip dhcp binding** and a client's `ipconfig` should always be cross-checked together — and re-pulled close together in time — to confirm a lease actually matches what the server thinks it handed out.

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*Compiled from the Suren Network lab screenshots (topology + 11 CLI/ipconfig captures) for CCNA self-study.*